Part A

$$U \leq n(2^{1/n} - 1)$$

$$U = C/T$$

set 1 : $\tau_1 = (T = 10,\ C = 2) \cdot \tau_2 (T = 25,\ C = 5) \cdot \tau_3 (T = 100, 20)$

$U_{\tau_1} = \frac{2}{10} = 0.20$ $\quad\quad$ * $n = 3$

$U_{\tau_2} = \frac{5}{25} = 0.20$ $\quad\quad$ bound $= 3(2^{1/3} - 1)$

$U_{\tau_3} = \frac{20}{100} = 0.20$ $\quad\quad$ $= 0.77$

$U_T = 0.60$

$U_T \leq 0.77$ $\therefore$ feasible for RMS

$U_T \leq 1$ $\therefore$ feasible EDF

set 2 : $\tau_1 (T = 4,\ C = 1) \cdot \tau_2 (T = 6,\ C = 2) \cdot \tau_3 (T = 10,\ C = 3)$

$U_{\tau_1} = \frac{1}{4} = 0.25$ $\quad\quad$ * $n = 3$

$U_{\tau_2} = \frac{2}{6} = 0.3\bar{3}$ $\quad\quad$ bound $= 0.77$

$U_{\tau_3} = \frac{3}{10} = 0.30$

$U_T = 0.88$

$U_T \geq 0.77$ $\therefore$ RMS is NOT feasible

$U_T \leq 1$ $\therefore$ EDF is feasible

set 3 : $\tau_1 (T = 5,\ C = 1) \cdot \tau_2 (T = 10,\ C = 2) \cdot \tau_3 (T = 20,\ C = 4) \cdot \tau_4 (T = 50,\ C = 10)$

$U_{\tau_1} = \frac{1}{5} = 0.20$ $\quad$ * $n = 4$

$U_{\tau_2} = \frac{2}{10} = 0.20$ $\quad$ $\text{bound} = 4(2^{1/4} - 1)$

$U_{\tau_3} = \frac{4}{20} = 0.20$ $\quad$ $= 0.75$

$U_{\tau_4} = \frac{10}{50} = 0.20$

$U_T = 0.80$

$U_T \geq 0.75 \therefore$ RMS is NOT Feasible

$U_T \leq 1 \therefore$ EDF is Feasible

Part B

set 1 : $\tau_1 = (T = 10, C = 2) \cdot \tau_2 (T = 25, C = 5) \cdot \tau_3 (T = 100, 20)$

shorter period → higher priority

$\tau_1 > \tau_2 > \tau_3$

Task 1 :

$R_1 = C \rightarrow 2\text{ms}$

$R_1 < T \rightarrow 2 < 10 \therefore$ Feasible

Task 2 :

$i_0 - R_2^{(0)} = C \rightarrow 5\text{ms}$

$i_1 - R_2^{(1)} = 5 + (\frac{5}{10} \cdot 2) \rightarrow 5 + (1 \cdot 2) = 7\text{ms}$

↳ round 0.5 up

$i_2 - R_2^{(2)} = 5 + (\frac{7}{10} \cdot 2) \rightarrow 5 + (1 \cdot 2) = 7\text{ms}$ [converged]

↳ round 0.1 up

$R_2 = 7ms$

$R_2 < T_2 \rightarrow 7 \leq 25 \therefore$ Feasible

Task 3:

$i_0 - R_3^{(0)} = C \rightarrow 20ms$

$i_1 - R_3^{(1)} = 20 + \left\lceil \frac{20}{10} \cdot 2 \right\rceil + \left\lceil \frac{20}{25} \cdot 5 \right\rceil = 20 + \lceil 2 \cdot 2 \rceil + \lceil 1 \cdot 5 \rceil = 29ms$

$i_2 - R_3^{(2)} = 20 + \left\lceil \frac{29}{10} \cdot 2 \right\rceil + \left\lceil \frac{29}{25} \cdot 5 \right\rceil = 20 + \lceil 3 \cdot 2 \rceil + \lceil 2 \cdot 5 \rceil = 36ms$

$i_3 - R_3^{(3)} = 20 + \left\lceil \frac{36}{10} \cdot 2 \right\rceil + \left\lceil \frac{36}{25} \cdot 5 \right\rceil = 20 + \lceil 4 \cdot 2 \rceil + \lceil 2 \cdot 5 \rceil = 38ms$

$i_4 - R_3^{(4)} = 20 + \left\lceil \frac{38}{10} \cdot 2 \right\rceil + \left\lceil \frac{38}{25} \cdot 5 \right\rceil = 20 + \lceil 4 \cdot 2 \rceil + \lceil 2 \cdot 5 \rceil = 38ms$ [converged]

$R_3 = 38ms$

$R_3 \leq T_3 \rightarrow 38 \leq 100 \therefore$ Feasible

Part C

Set 3: $\tau_1(T=5, C=1) \cdot \tau_2(T=10, C=2) \cdot \tau_3(T=20, C=4) \cdot \tau_4(T=50, C=25)$

$U_{\tau_1} = 0.20 \quad U_{\tau_2} = 0.20 \quad U_{\tau_3} = 0.20 \quad U_{\tau_4} = \frac{25}{50} = 0.50$; bound $= 0.75$

$U_T = 1.10$

NOT RMS Feasible because $U_T \geq 0.75$

NOT EDF Feasible because $U_T \geq 1$

Priority $\tau_1 > \tau_2 > \tau_3 > \tau_4$

Task 1 :

i_0 - $R_1 = C \rightarrow 1ms$

$R_1 < T_1 \rightarrow 1ms \leq 5 \therefore$ Feasible

Task 2 :

i_0 - $R_2 = C \rightarrow 2ms$

i_1 - $R_2^{(1)} = 2 + \frac{2}{5} \cdot 1 = 2 + 1 = 3ms$

i_2 = $R_2^{(2)} = 2 + \frac{3}{5} \cdot 1 = 2 + 1 = 3ms$ [converged]

$R_2 = 3ms$

$R_2 \leq T_2 \rightarrow 3 \leq 10 \therefore$ Feasible

RTA will confirm